

Assignment 1

Computer Networks

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Network Commands

COMMAND	SYNTAX	PURPOSE	OUTPUT
ping	<code>ping <hostname></code>	Tests network connectivity.	64 bytes from 142.250.183.206: icmp_seq=0 ttl=117 time=18.2 ms
tracert	<code>tracert <domain></code>	Shows the packet route to a destination.	1 192.168.1.1 → 2 ISP Router → ... → Destination
ipconfig	<code>ipconfig</code>	Displays network interface details.	inet 192.168.1.10 netmask 255.255.255.0 ether 00:1A:2B:3C:4D:5E
arp	<code>arp -a</code>	Displays the ARP cache.	? (192.168.1.1) at 00:11:22:33:44:55 on en0
netstat	<code>netstat</code>	Shows network connections and ports.	tcp4 0 0 192.168.1.10.52134 142.250.183.206.443 ESTABLISHED
nslookup	<code>nslookup <domain></code>	Resolves domain names to IP addresses.	Name: google.com Address: 142.250.183.206
route	<code>route -n get default</code>	Displays routing information.	gateway: 192.168.1.1 interface: en0
nmap	<code>nmap localhost</code>	Scans for open ports and services.	22/tcp open ssh80/tcp open http
pathping	<code>pathping <hostname></code>	Analyzes network path and packet loss.	Hop 1 192.168.1.1 0% loss 1 ms

nbtstat	nbtstat [options]	Displays NetBIOS information.	NetBIOS Local Name Table
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1. ping

Purpose:

The **ping** command is used to test the connectivity between the local computer and another host on the network. It also measures the time taken for packets to travel to the destination and back.

Syntax:

```
ping <hostname/IP address>
```

Output:

```
[kavyabhand@Kavyas-MacBook-Air-2 ~ % ping google.com
PING google.com (142.250.134.101): 56 data bytes
64 bytes from 142.250.134.101: icmp_seq=0 ttl=116 time=11.459 ms
64 bytes from 142.250.134.101: icmp_seq=1 ttl=116 time=10.530 ms
64 bytes from 142.250.134.101: icmp_seq=2 ttl=116 time=12.673 ms
64 bytes from 142.250.134.101: icmp_seq=3 ttl=116 time=10.222 ms
^C
--- google.com ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 10.222/11.221/12.673/0.954 ms
```

2. traceroute / tracert

Purpose:

The **traceroute** command displays the path followed by packets from the source computer to the destination host through intermediate routers.

Syntax:

```
traceroute <domain>
```

Output:

```
[kavyabhand@Kavyas-MacBook-Air-2 ~ % traceroute google.com
traceroute: Warning: google.com has multiple addresses; using 192.178.211.101
traceroute to google.com (192.178.211.101), 64 hops max, 40 byte packets
 1  192.168.1.254 (192.168.1.254)  4.006 ms  3.341 ms  2.988 ms
 2  * * *
 3  * * ^C
```

3. ifconfig / ipconfig

Purpose:

The **ifconfig** command displays and configures network interface information such as IP address, MAC address, subnet mask, and network status.

Syntax:

```
ifconfig
```

Ouput:

```
[kavyabhand@Kavyas-MacBook-Air-2 ~ % ifconfig
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> mtu 16384
    options=1203<RXCSUM,TXCSUM,TXSTATUS,SW_TIMESTAMP>
    inet 127.0.0.1 netmask 0xff000000
    inet6 ::1 prefixlen 128
    inet6 fe80::1%lo0 prefixlen 64 scopeid 0x1
    nd6 options=201<PERFORMNUD,DAD>
gif0: flags=8010<POINTOPOINT,MULTICAST> mtu 1280
stf0: flags=0<> mtu 1280
anpi1: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether 46:cc:1c:da:3a:5f
    media: none
    status: inactive
anpi0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether 46:cc:1c:da:3a:5e
    media: none
    status: inactive
en3: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether 46:cc:1c:da:3a:3e
    nd6 options=201<PERFORMNUD,DAD>
    media: none
    status: inactive
en4: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether 46:cc:1c:da:3a:3f
    nd6 options=201<PERFORMNUD,DAD>
    media: none
    status: inactive
```

4. arp

Purpose:

The **arp** command displays and manages the Address Resolution Protocol (ARP) cache, which maps IP addresses to MAC addresses.

Syntax:

```
arp -a
```

Output:

```
[kavyabhand@Kavyas-MacBook-Air-2 ~ % arp -a
? (192.168.1.4) at c2:6:c3:2:96:2 on en0 ifscope [ethernet]
? (192.168.1.10) at b6:9d:d3:be:2:50 on en0 ifscope [ethernet]
? (192.168.1.14) at 8e:3f:9b:2:2c:88 on en0 ifscope [ethernet]
? (192.168.1.254) at 54:37:bb:67:a6:50 on en0 ifscope [ethernet]
? (192.168.1.255) at ff:ff:ff:ff:ff:ff on en0 ifscope [ethernet]
mdns.mcast.net (224.0.0.251) at 1:0:5e:0:0:fb on en0 ifscope permanent [ethernet]
? (239.255.255.250) at 1:0:5e:7f:ff:fa on en0 ifscope permanent [ethernet]
```

5. netstat

Purpose:

The **netstat** command displays active network connections, listening ports, interface statistics, and routing information.

Syntax:

```
netstat
```

Output:

```
[kavyabhand@Kavyas-MacBook-Air-2 ~ % netstat
Active Internet connections
Proto Recv-Q Send-Q Local Address Foreign Address (state)
tcp6 0 0 2402:e280:3e18:2.49829 whatsapp-cdn6-sh.https ESTABLISHED
tcp4 0 0 192.168.1.9.49828 ec2-44-215-234-2.https ESTABLISHED
tcp4 0 0 192.168.1.9.49827 s3-us-west-2-r-w.https CLOSE_WAIT
tcp4 24 0 192.168.1.9.49826 s3-us-west-2-r-w.https CLOSE_WAIT
tcp6 0 0 2402:e280:3e18:2.49825 bom12s05-in-x05..https ESTABLISHED
tcp6 0 0 2402:e280:3e18:2.49824 lcbomp-in-f84.1e.https ESTABLISHED
tcp4 0 0 192.168.1.9.49823 server-18-161-12.https ESTABLISHED
tcp6 0 0 2402:e280:3e18:2.49811 2606:4700:4407::.https ESTABLISHED
tcp6 0 0 2402:e280:3e18:2.49809 2606:4700::6812:.https ESTABLISHED
tcp6 0 0 2402:e280:3e18:2.49808 2606:4700::6812:.https ESTABLISHED
tcp6 0 0 2402:e280:3e18:2.49776 2602:f79a::1.https ESTABLISHED
tcp4 0 0 192.168.1.9.49774 207.164.110.34.b.https ESTABLISHED
tcp6 0 0 2402:e280:3e18:2.49760 2606:4700:440b::.https ESTABLISHED
tcp6 0 0 2402:e280:3e18:2.49758 2606:4700:4408::.https ESTABLISHED
tcp6 0 0 2402:e280:3e18:2.49751 2600:1901:0:5e8a.https ESTABLISHED
tcp4 0 0 192.168.1.9.49729 23.109.134.67.https ESTABLISHED
tcp6 0 0 2402:e280:3e18:2.49713 2606:4700::6812:.https ESTABLISHED
tcp4 0 0 192.168.1.9.49711 202.241.199.104..https ESTABLISHED
tcp6 0 0 2402:e280:3e18:2.49322 lcbomp-in-f188.1.5228 ESTABLISHED
tcp6 0 0 2402:e280:3e18:2.49318 lcbomp-in-f188.1.5228 ESTABLISHED
tcp6 0 0 2402:e280:3e18:2.49256 2600:1901:1:d18:.https ESTABLISHED
tcp6 0 0 2402:e280:3e18:2.57654 dns.google.443 ESTABLISHED
tcp4 0 0 192.168.1.9.53654 17.242.13.6.5223 ESTABLISHED
tcp6 0 0 2402:e280:3e18:2.57520 2600:1901:1:d18:.443 ESTABLISHED
```

6. nslookup

Purpose:

The **nslookup** command is used to obtain the IP address associated with a domain name using the Domain Name System (DNS).

Syntax:

```
nslookup <domain-name>
```

Output:

```
[kavyabhand@Kavyas-MacBook-Air-2 ~ % nslookup google.com
Server:      2001:4860:4860::8888
Address:     2001:4860:4860::8888#53

Non-authoritative answer:
Name:   google.com
Address: 192.178.173.100
Name:   google.com
Address: 192.178.173.138
Name:   google.com
Address: 192.178.173.139
Name:   google.com
Address: 192.178.173.113
Name:   google.com
Address: 192.178.173.101
Name:   google.com
Address: 192.178.173.102
```

7. route

Purpose:

The **route** command displays the routing table and the default gateway used by the operating system for forwarding network packets.

Syntax:

```
route -n get default
```

Output:

```
[kavyabhand@Kavyas-MacBook-Air-2 ~ % route -n get default
route to: default
destination: default
mask: default
gateway: 192.168.1.254
interface: en0
flags: <UP,GATEWAY,DONE,STATIC,PRCLONING,GLOBAL>
recvpipe  sendpipe  ssthresh  rtt,msec  rttvar  hopcount  mtu  expire
0          0          0          0          0        0         1500  0
```

8. nmap

Purpose:

The **nmap** command is used to scan hosts and identify open ports, running services, and network information.

Syntax:

```
nmap localhost
```

Output:

```
[kavyabhand@Kavyas-MacBook-Air-2 ~ % nmap localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-07 09:00 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000037s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 995 closed tcp ports (conn-refused)
PORT      STATE SERVICE
3306/tcp  open  mysql
5000/tcp  open  upnp
5432/tcp  open  postgresql
7000/tcp  open  afs3-fileserver
49152/tcp open  unknown

Nmap done: 1 IP address (1 host up) scanned in 0.05 seconds
```

9. pathping

Purpose:

The **pathping** command combines the features of **ping** and **tracert** to identify packet loss and network latency at each hop.

Syntax:

```
pathping <hostname/IP address>
```

Output:

```
tracert: Warning: google.com has multiple addresses; using 142.250.134.138
tracert to google.com (142.250.134.138), 64 hops max, 40 byte packets
 1  192.168.1.254 (192.168.1.254)  5.382 ms  3.207 ms  3.342 ms
 2  * * *
 3  * * *
 4  72.14.198.2 (72.14.198.2)  38.404 ms  39.398 ms  41.790 ms
 5  192.178.110.125 (192.178.110.125)  43.474 ms
    192.178.110.223 (192.178.110.223)  13.001 ms
    192.178.99.151 (192.178.99.151)  11.980 ms
 6  192.178.110.206 (192.178.110.206)  9.189 ms
    192.178.110.198 (192.178.110.198)  9.567 ms
    172.253.177.90 (172.253.177.90)  11.873 ms
 7  142.251.78.223 (142.251.78.223)  12.637 ms
    64.233.175.15 (64.233.175.15)  14.770 ms
    142.251.230.107 (142.251.230.107)  11.449 ms
```

10. nbtstat

Purpose:

The **nbtstat** command is used to display **NetBIOS over TCP/IP** statistics, NetBIOS name tables, and NetBIOS name cache information.

Syntax:

```
nbtstat [options]
```